
CS 406: Computer Networks

Course Code	CS 406
Course Name	Computer Networks
L-T-P-C	3-1-0-4
Prerequisites	CS-304 - Communicating Distributed Processes or the instructor's consent
Intended for	UG
Distribution	Elective for CS and EE
Semester	7th or 8th
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to Networking and Physical layer** [4 Lectures]
 - Signals and Data Basic Properties, Fourier Transform, Transmission (noise, attenuation, distortion), Effective Bandwidth.
 - Digital to Digital Transmission Line coding, Block coding and Scrambling.
 - Multiplexing - Frequency-Division Multiplexing (FDM), Wavelength-Division Multiplexing (WDM), Time-division Multiplexing (TDM).
 - Types of Switching Circuit Switching, Packet Switching, Message Switching, QoS for Circuit Switching.
- **2. Data Link Layer** [7 Lectures]
 - Link-Layer Addressing Types of addresses, Address Resolution Protocol (ARP).
 - Error Detection and Correction Block Coding, Cyclic Codes (Cyclic Redundancy Check, Polynomials, GF2) and Checksum.
 - Logical Link Control (LLC) Sublayer: LLC Services Framing, Flow and Error Control. Protocols Stop-and-Wait, Piggybacking, High-Level Data Link Control (HDLC), Point-to-Point Protocol (PPP).
 - Media Access Control (MAC) Sublayer: Random Access ALOHA, Carrier Sense Multiple Access (CSMA), Collision Detection (CSMA/CD), Collision Avoidance (CSMA/CA).
- **Network Layer** [10 Lectures]
 - Network Layer Services Packeting, Routing, and Forwarding.
 - Packet Switching Datagram Approach (connectionless Service), Virtual Circuit Approach (connection- oriented Service).
 - QoS of Packet Switching Delay, Throughput, Packet Loss, and Congestion Control.
 - IP Addressing IPv4 and IPv6, Dynamic Host Configuration Protocol (DHCP), Network Address Translation (NAT).
 - Routing Algorithms:
 - * UNICAST Routing: Max Flow-Min Cut theorem and Ford Fulkerson Algorithm, Bellman Ford and Dijkstra's Algorithms, Shortest Path Routing, Distance-Vector Routing, Link -State Routing and Path Vector Routing, Hierarchical Routing, Routing Information Protocol (RIP), Open Shortest Path First (OSPF).
 - * MULTICAST Routing: Flooding, Multicast Distance Vector (DVMRP), Protocol Independent Multicast (PIM) and Multicast Link State (MOSPF).

- **Transport Layer** [10 Lectures]
 - Protocols Stop-and-Wait, Go-Back-N, Selective-Repeat, User Datagram Protocol (UDP): services and applications, Transmission Control Protocol (TCP): services, flow, and congestion control.
 - TCP implementation techniques Slow start, congestion avoidance, fast retransmit and fast recovery.
 - Optional Topics: TCP Variants: TCP Tahoe, TCP Reno, TCP Vegas, TCP SACK and TCP Westwood, STCP.
- **Application Layer** [6 Lectures]
 - Services and Protocols Application protocols, such as Hypertext Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Internet Mail Access Protocol (IMAP), Post Office Protocol (POP), Simple Network Management Protocol (SNMP).
 - Optional Topics: Secure Shell (SSH), Domain Name System (DNS), Lightweight Directory Access Protocol (LDAP), Network Time Protocol (NTP), Remote Procedure Call (RPC) and Secure Socket Layer (SSL).
- **Advanced Topics** [5 Lectures]
 - Service Models Resource Reservation Protocol (RSVP), Integrated and Differentiated.
 - Flow Control Scheduling, Traffic Modellers, Multi Protocol Label Switching (MPLS).
 - Peer to Peer Paradigm Distributed Hash Table (DHT), CHORD, PASTRY, KADEMILA.

Text Books

1. B. A. Forouzan, **Data Communications and Networking**, 5th Edition, McGraw Hill, 2013.

References

1. A. S. Tanenbaum and D. J. Wetherall, **Computer Networks**, 5th Edition, Prentice Hall, 2011.
2. D. P. Bertsekas and R. G. Gallager, **Data Networks**, 2nd Edition, Prentice Hall, 1992.
3. S. Keshav, **An Engineering Approach to Computer Networking: ATM Networks, the Internet, and the telephone network**, Addison Wesley Longman, 1997.
4. R. Perlman, **Interconnections: Bridges, Routers, Switches and Internetworking Protocols**, 2nd Edition, Addison Wesley, 1999.
5. L. L. Peterson and B. S. Davie, **Computer Networks: A Systems Approach**, 5th Edition, Morgan Kaufmann, Elsevier, 2011.
6. W. Stallings, **Data and Computer Communications**, 10th Edition Pearson, 2013.
7. G. B. White, E. A. Fisch, and U. W. Pooch, **Computer System and Network Security**, CRC Press, 1995.